CPT312 - PLANT PHYSIOLOGY

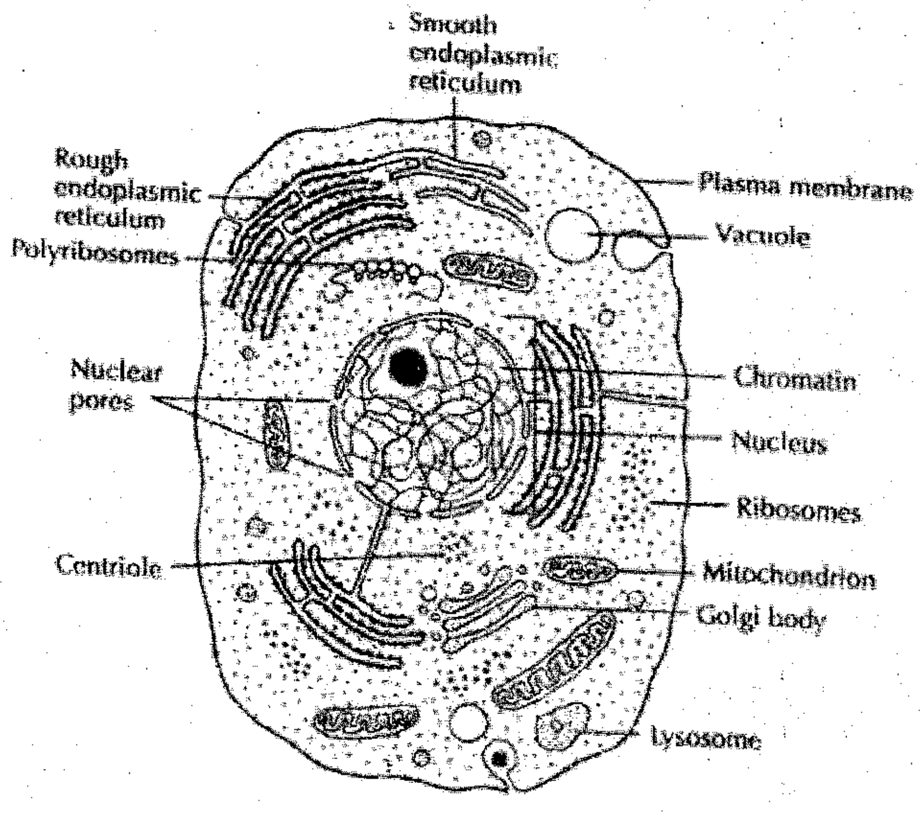
CELL

Definition:

The cell is the fundamental morphological and physiological unit of living organisms. The cell contains all the structures and molecular constituents needed for life. Cell of a living organism is divided into two groups: (1) Unicellular (2) Multicellular organisms.

1. Unicellular organisms are those which start their life from a single cell unit and till death continue in the same manner; their body is made up of a single cell. E.g. yeast, diatoms bacteria and protozoans.
2. Multicellular organisims are those which are made up of more than one cell these organisms also start their life from a single cell but later on it undergoes continued division and develop in to a multicellular body all higher plans and animals are multicellular. The cell is multicellulaer organisms get arranged in a definite manner to form different parts of the body. These parts perform different functions and all are governed by the common activity of each cell present in the part of the body.

   Electro microscopic studies have revealed that each cell contains in it several living sub-cellular particles called cell-organelles which perform different functions, the chief cell-organisms are mitochondria, endoplasmic reticulum, ribosome, plastids, lysosome, golgi body and nucleus e.g. this it can be said that despite being the unit of living organisms, cell has a much complex structure.



Types of cell

1. Prokaryotic cell
2. Eukaryotic cell

1. Prokaryotic cell (pro = primitive, karyon = nucleus). They are characterized by the absence of nuclear membrane, nucleus, nucleous and most of the well developed cytoplasmic organelles. They are been considered the most primitive type of cells and in them nuclear material is found freely distributed in the cytoplasm, i e. the nuclear materials like DNA, RNA and proteins are in direct contain with protoplasm. The chromosome of prokaryotic cell lack histone proteins. Bacteria, viruses, rickettsia, bacteriophages and blue green algae are examples of prokaryotic cell.
2. Eukaryotic cell (greek Eu-good or well, karyon-nucleus). Eukaryotic cells possess all the characters which prokaryotic cells lack. Thus, they are characterized by the present of definitely organized nucleus with nuclear cytoplasm organelles like mitochondria, plastids, ribosomes, endoplasmic reticulum, lysosomes, golgi body etc. nuclear materials like DNA, RNA and proteins in these cell are in contact with karyoplasms. Their chromosomes contain histone protein. E.g algae (except blue green algae) and all higher plant and animal cell.

Osmotic relations of plant cells

Diffusion: The movement of free molecules of gases, liquids and solid from the region of higher concentration to the region of lower concentration due to internal or external forces is called diffusion. Blowing of wind dispersal of good smell of agarbattis in a room, dissolution of sugar in water, isntake of CO2 and libration of O2 in photosynthesis, in take of O2 and libration of O_2 during respiration are all the example of diffusion. Such a diffusion of molecules in any region contains it spreads throughout the available area. The movement of molecules depends upon the internal kinetic energy. The molecules in the region of higher concentration contain more kinetic energy and that is why they slow fast movement. During the movement all the molecules collide themselves and produce a pressure in the medium called diffusion pressure which is directly proportional to the concentration and temperature of diffusing molecules. It means if the temperature and concentration is loss, the diffusion pressure will also be use.

1. Diffusion pressure and concentration of molecules

2. Diffusion pressure and temperature.

Due to this reason, the diffusion always takes place from the region of higher diffusion pressure (conc.) to the region of lower diffusion pressure.

Mostly gases, liquid and solutes diffuse at different rates in different direction at the same place without affecting each other. This is the reason that O_2 and CO_2 both diffuse in different direction from a single stomata of leaf during respiration. The rate of diffusion is maximum in gases and minimum in solids when dissolved in solvent liquids stand on intermediate position in this regard.

Factors affecting the rate of diffusion

1. Temperature: It is directly proportional to the rate of diffusion. On increasing the temperature, the rate of diffusing particles is also increased because of increase in the velocity of these particles.
2. Concentration of medium: It is inversely proportional to the rate of diffusion. On increasing the concentration of the medium, the rate of diffusion is reduced and on decreasing the concentration, it is increase.
3. The size and mass of the diffusion particles: If the size and mass of the diffusion particle is smaller, the rate their diffusion will always be faster.
4. Solubility of solutes: The rate of diffusion increase with the rate of dissolution of solute in solution. Thus, the more the solubility of any solute in solution, the faster will be the rate of diffusion.
5. Diffusion pressure gradient (DPG) : The rate of diffusion molecules of gases and liquid also depends upon the diffusion pressure gradient (DPG). When the difference in the diffusion pressure is more, faster is the rate of diffusion.
6. Density of the diffusion particles: The rate of diffusion of gases is related with the density of diffusion particles. According to Grahm's law of diffusion, " the rate of diffusion of any gas particle is inversely proportional to the square root of its density.

$r = \frac{1}{\sqrt{d}}$ Where: r = rate of diffusion of a gas
d = density of the gas

The gases having higher densities shows slower rate of diffusion where as those possessing lower density slower faster rate of diffusion. For example, the density of hydrogen (H) is one and of oxygen (O) 16. Therefore, according to Grahm's low of diffusion:

$$\frac{rH}{rO} = \frac{\sqrt{do}}{\sqrt{dH}} = \frac{\sqrt{16}}{\sqrt{1}} = \frac{4}{1}$$

Thus, the hydrogen will diffuse four times faster than oxygen.

Importance of diffusion in plants

i. The exchange of CO_2 gates in the atmosphere through stomata of leaves takes place by the process of diffusion. O_2 gas participates in respiration whereas CO_2 in photosynthesis.

ii. During stomata transpiration the water vapors from intercellular spaces diffuse in the atmosphere through stomata by the process of diffusion.

iii. The diffusion of ions of mineral salt during passive absorption takes by this process.

iv. The absorption of water through roots is also performed by diffusion.

Permeability: The entrance of exit of any substance in the cell depends upon the permeability of plasma membrane. The permeability of plasma membrane is usually changes from times to time due to changes in either K^+ ions concentration of nature of other particles surrounding it. Depending upon the permeability properties the membranes can be classified into following categories.

(a) Impermeable: these are those membranes through which of only the following exchanges of only gates takes place.

(b) Semi-permeable: these are permeable through which exchange of only water or solvent molecules takes place. Solute particles neither can enter nor leave through of goat, and egg membranes are the examples of semi-permeable membrane.

(c) Selective permeable: These are the membranes through which the exchange of only selected ions or small molecules takes place. Plasma membrane is mainly selective permeable and not only semi-permeable.

(d) Dialyzing membranes; These membranes possess other layers in the form of water covering.

Plasma membrane is usually permeable for gases like CO_2, O_2, N_2, solvents like alcohol, either chloroform and organic substance like mono- and disaccharides, fatty acids, amino acids and glycerol whereas it is normally impermeable for polysaccharides and proteins. The permeability of plasma membrane for gases and solvent is more than organic substances.

Factors affecting permeability

A number of external and internal factors affect the permeability of plasma membrane.

External factors

1. Physical agents: the physical agents like high temperature, heat, low pressure of O_2 and CO_2, radiation etc. increase the permeability of plasma membrane.
2. Chemical agents: a number of chemicals like ether, benzene, chloroform, acetone, toxic substances etc. when added in the external medium or solution increase the permeability
3. Ageing of cells: The permeability various at different ages of the cell. In young cells, it is how but at the times of senescence it increases the older cells approaching death also shows increase in permeability.

Internal factors

V Membrane Constitution: the permeability of plasma membrane depends upon its constitution, i.e. percentage of lipids, proteins, carbohydrates and phosphate etc. thickness, porosity and degree of hydration.

Osmosis:

When two solutions of different concentrations are separated by semi-permeable membrane, the diffusion of water or solvent molecules takes place from the solution of lower concentration towards the solution of higher concentration. This process is called osmosis.

Actually the diffusion of solvent molecule takes place across the membrane on both the sides but it is faster from the lower concentration side than the higher concentration side. This is according to principles of diffusion because the solution of lower concentration possesses higher concentration of solvent molecule whereas the solution of higher concentration possesses comparatively lower osmosis is a type of diffusion through semi-permeable membrane.

From biological point of view, the solutions may be of following three types:

1. Hypertonic solutions: those solutions whose concentration and osmotic pressure (OP) are more than the concentration and OP of cell-sap are called hypertonic solutions.
2. Hypotonic solutions: Those solutions whose concentrations and OP are less than the concentration and OP and cell-sap, are called hypotonic solutions.
3. Isotonic solutions: Those solutions, whose concentration and OP are equal to the concentration and OP of cell-sap, are called Isotonic Solutions.

Types of Osmosis: osmosis in plants is of two types:

i. Endosmosis: when water or solvent molecules enter into the cell through plasma membrane from the outer medium, it is called endosmosis.

ii. Exosmosis when a plant cell is placed in concentrated solution, the water molecules move from cell into the outer concentration medium through plasma membrane. It is called exosmosis. Thus, in exosmosis the water or solvent molecules leaves the cell into the outer medium through plasma membrane.

Significance of osmosis in plants

1. The water is most important factor for life because all the biological activities are directly or indirectly controlled by water. Osmosis helps the plant in the absorption of water from the soil through roots.
2. The turgidity in plant cell is maintained by the absorption of water through osmosis.
3. The diffusion of water from one cell to another and its distribution in different plant parts is also performed by osmosis.
4. The opening of stomata depends upon the turgidity of guard cell which is maintained by osmosis.
5. Osmosis pressure and turgidity or pressure helps in the growth of young cells.
6. High osmosis concentration in plants increases the resistance against freezing temperature and dehydration.
7. Osmosis helps in the dehiscence of fruits and sporangia.

Differences between diffusion and osmosis

S/N	Diffusion	Osmosis
1	Diffusion occurs in solid, liquid and gases	Osmosis occurs only in liquid
2	Does not require semi permeable membrane	Requires a semi permeable membrane
3	In this process, the movement of molecules takes place from higher concentration	The movement of water or solvent molecules takes place from the solution of lower concentration to the solution of higher concentration

Abortion of water

Mechanism of water absorption

According tom Renner (1912 and 1915), the mechanism of water absorption is of following two types:

1. Active absorption: when roots are involved actively in water absorption and water absorbing forces in plants are developed primarily in roots, such types of absorption are called active absorption. It is found in those plants where transpiration is less and water is present in sufficient amounts. Active absorption requires ATP released during respiration. It is also of two types:
   (a) Osmotic absorption: in this type of absorption the roots act like osmometer and water absorption according to osmotic gradicut.
   (b) Non-osmotic absorption: in this type of water absorption more ATP is required and water absorption takes place against osmotic gradient.
2. Passive absorption: when roots are in active in water absorption and the water absorption forces develop primarily in leaves and stem and then reach to root through xylem, this types of water absorption is called passive absorption. It takes place mainly due to transpiration. Passive absorption is found in those plants where transpiration is very fast and it does not require ATP.
3. Osmosis active absorption: soil is made up of irregularly arranged soil particles. Well developed air spaces are found among those particles. The spaces contain capillary water where air and mineral salts remain dissolved. The roots having remain spread in this water. it was already been described that each root hair contains a vacuole which remains filled with mineral salts, sugar and organic acids. The cell-wall of root hair is permeable and it does not create any hindrance in the entry and exit of the liquids in the cell. Plasma membrane is semi permeable and it allows only the different. The cell-wall of root hair being hydrophilic in nature first absorbs soil water through ion bibityion. The cytoplasm of root hair is usually concentrated than the capillary water of soil.

Pathway of water in root

The pathway of water in root cell is as follows:

Root hair → Epidermis cell → various successive → cortex cell →
Endodermal cell (passage cell) → cell of pericycle → xylem cell →
Xylem duct → upward movement of water.

ROOT PRESSURE: when the water enters from turgid pericycle cell into xylem vessels, a pressure is created in the xylem of roots due to which the water rises to a certain height in the xylem. This pressure is called root pressure. Actually the root pressure is a type of hydrostatic pressure which is produced in the cell-sap of xylem vessels.

Theories regarding development of root pressure

Following theories have been proposed to explain how the root pressure develops in plants;

1. Secretion theory: according to this theory. The root pressure develops by the secretion of water into the xylem. The secretion of water results due to difference in the permeabilities of water in two sides of the root cells. It is higher towards inner side and lower towards outer side. Some energy may be required for this work which is provided by respiration through root cell.
2. Electro-osmotic theory: according to this theory the root pressure develops due to electro-osmosis movement of water into the xylem. It has been proved experimentally that when an electric current is passed into the membrane, water moves across the membrane. However, this theory was seen discarded.
3. Osmotic theory: according to this theory the root pressure develops due to osmosis. The root behaves like an osmometer due to accumulation of xylem gap. Thus, water moves from soil solution to xylem through various tissues of root developing root pressure.

Non-osmotic active absorption

According to Thimann (1951) and Kramer (1959) the water absorption is an active process which takes place against the osmotic gradient. In other words, sometimes the water absorption also takes place when the osmotic pressure of soil water, is greater than the osmotic pressure of cytoplasm of root hair. Thus is against osmosis. Such type of water absorption is called non-osmotis active absorption and it requires ATP produced during respiration of root cells. How this energy is utilized, it is not well worked out. The energy may be utilized directly, or indirectly. The theory of non-osmotic active absorption is supported by the following facts.

i. Correlation between water absorption and rate of respiration usually the presence of respiration reducing factors the rate of water absorption is also reduced. This, both the process inter-related

ii. Reduction of water absorption in presence of respiratory inhibitions: it has been observed when plant is placed in the solution of respiratory inhibitions like KCN, the rate of water secretion is also reduced or stopped. This indicates that respiration and water absorption processes are inter-related. KCN directly, destroys the coenzymes of respiratory chain like NAD, FAD and cyto-chromes and thus effects respiration and other metabolic activities

iii. Wilting of plants in oxygen deficient soil: in O_2 deficient soil, the plants do not get sufficient O_2 for respiration and ultimately unit

iv. Effects of auxins: in presence of auzins, the rate of metabolisms along with water absorption are increased.

v. The occurrence of water absorption and respiration only in living cells. Both these processes are found only in living plants and not in dead plant.

Differences between active and passive absorption

S/N	Active absorption	Passive absorption
1	It requires or ATP	It does not require energy
2	It creates root pressure	Root pressure is not created due to P.A
3	It requires oxygen	It does not requires oxygen
4	The movement of higher concentration of the solution of lower concentration i.e. against the concentration gradient	The movement of water takes place from the solution of lower concentration to the solution of higher concentration i.e. according to osmotic gradient

Factors affecting the rate of water absorption

1. Available soil water: It has been discovered that soil water is found in many forms (gravitational, Capillary, Hygroscopic and chemically bound water) but out of these only capillary water is available for the absorption of plants. The rate of water absorption in plants remains constant the same in between field capacity and permanent wilting percentage. If the amount of water is increased than the fields capacity, it creates a bad effect on soil aeration and also reduces the rate of water absorption. Similarly, on shortage, the water absorption rate is also reduced.
2. Soil aeration: The rate of water absorption increases when the aeration in soil is high. In other words, the rate of water absorption is increased is due to lack of O_2 and accumulation of CO_2 which effect the plants in following manner.
   i. CO_2 reduces metabolism activities and respiration rates
   ii. It reduces the size and growth of roots
   iii. It increases the concentration of protoplasm which result in the reduction of permeability.

Due to above effects of CO_2, the rate of water absorption is decreased. This situation is created in water-logged soil where the aeration is quite poor. Such soil is usually physiologically dry and is not fit for water absorption.

3. Concentration of the soil solution: the osmotic pressure of any solution is directly proportional to its concentration. In other words, the OP is more concentrated solutions. If the soil water containing more salts, its concentration and OP will also be more. The rate of water absorption will be reduced. If the OP of soil solution will be more than OP of cell sap of roots. Due to this reason, the plants growing in alkaline and marsh soil shows a very little or no water absorption.
4. Soil temperature: generally, the rate of water absorption is maximum between 20-30^0C. it is reduced if the temperature is less than 20^0C or more than 30^0C. at very low temperature or at 0^0C, the water absorption is almost stopped and its rate of water absorption is affected directly or indirectly in the following manners:

i. It increases the concentration of protoplasm
ii. It reduces the permeability of plasma membrane
iii. The growth of root is reduced which result in reduction of their length
iv. The rate of diffusion of soil-water into roots is also reduced
v. Metabolic activities of root cells are also reduced

5. Root system: root system also affects the rate of water absorption. Those plants which possess hairy and well developed root system flows higher rates of water absorption in comparison to those possess very small roots and fewer members of root hairs.

Importance of water to the plants

Although water plays many roles in plants but some of the important ones are:

1. The amount of water present in the soil changes the morphology and anatomy of the plants. Accordingly the plants are mesophytes, hydrophytes and xerophytes.
2. Water is good solvent for many substances such as organic salts, sugars and organic anions i e it dissolves other substances.
3. Water helps in the formation of protoplasm.
4. The absorption and translocation of mineral salt and dissolved substances take place through water.
5. Water is an essential raw material in photosynthesis. During photosynthesis, water is oxidized and O_2 is produced
6. Water affects transpiration seed germination, respiration, dispersal of fruits and seeds, activation of enzymes, hydrolysis of ATP growth and all other metabolic processes.

7. Water maintain the temperature in plant tissues
8. Water maintains the turgidity of plant body

Soil water

Soil water is the most important constituent of soil because it provides the medium of the absorption of minerals and organic matter by the roots and also activities various enzymes and metabolic processes. It affects morphological and physiological processes directly or indirectly. Soil-water later on reaches to the various other parts of the plant through roots. The vegetation of an area depends upon the quality of soil water. The water present in the soil may be of following types:

1. Gravitational water: The water which reaches deeply into the soil after rains due to gravitation is called gravitational water. The plants cannot absorb this water through their roots.
2. Surface tension: Is the forces which exists as a combination of cohesive force between water molecules, and adhesive force between water molecules and sand particles. It is this force which makes water drops spread on the surface of soil particles. It also enables water to rise up through narrow pores.
3. Capillary water: Is water which rises above the water table in the soil and is held in the fine and medium pores of soil particles by surface tension. Capillary water moves from more saturated to less saturated regions. The finer the pores, the greater is the force binding the water (i,e surface tension), and the higher is the capillary tension. This results in a higher rise of water. Capillary water is the only available water which is absorbed by plant roots.
4. Hygroscopic water: Is that water which is absorbed on the surfaces of soil particles from atmospheric water vapour. The colloidal components of the soil, such as clay and humus, have a strong attraction for water molecules, binding them to their surfaces by surface energy forces. The amount of water absorbed from the atmosphere is always in equilibrium with atmospheric moisture. It is non-available water and is not absorbed by plant roots.
5. Crystalline water or chemically combined water: The water which remains chemically bound to the soil particles is called crystalline water. It is also non-available water, e.g. $CuSo_4$, $5H_2O$, $FeSo_4$, $7H_2O$, etc.
6. Running water: The non-available water which after rains flow down through the slopes is called running water.